

Coding Foundations Passport

One planning record for blocks, games, text code, and robots.

Alias / initials:

Class:

Date started:

Set up this passport once

- Make one copy only. Rename it ClassPeriod_AssignedAlias_Coding_Passport and place it in your course coding folder. Use only a teacher-approved alias or initials—not your full name.
- Open this same copy every time. Do not make a new copy for each checkpoint.
- Complete only the checkpoint your teacher assigns. A checkpoint may be on more than one page.
- When you submit, capture the completed checkpoint page or section—not the whole passport.
- If your copy is lost, search the exact filename first. Then ask your teacher for the recovery route and use earlier Canvas submissions to restore prior evidence.

Remember

A screenshot shows that code ran. Your plan, test record, and explanation show what you understand.

Checkpoint 1 - Decompose and plan before blocks

Use with the shared Code.org route or the task your teacher names.

Goal: What must the finished program make happen?

Inputs and outputs: What enters the system? What should the system produce?

Three subproblems: Break the goal into three smaller jobs.

First pseudocode draft

```
START
  [first precise command]
  [next command]
  [repeat or decision if needed]
END
```

Partner literal test: Where did your partner stop because the plan was unclear or incomplete? What line will you revise?

Checkpoint 2 - Find a common pattern and generalize it

Use this common route example even if your Hour of Code activity uses different blocks.

Common route

```
MOVE forward 3 spaces
TURN right
MOVE forward 3 spaces
TURN right
MOVE forward 3 spaces
TURN right
MOVE forward 3 spaces
TURN right
```

Repeated group: Bracket the smallest useful group. Which two commands repeat?

Iteration benefit: Why is a loop better than copying these commands four times?

Common variable	Type	Starting value	What it controls / operation
sideLength	number	3	
turnDirection	string	"right"	
keepDrawing	Boolean	true	

Generalized pseudocode

```
SET sideLength TO 3
SET turnDirection TO right
REPEAT 4 times
  MOVE sideLength spaces
  TURN turnDirection
END REPEAT
```

Transfer: Name one pattern from today's coding activity. How is it similar to or different from the common route?

Checkpoint 3 - Predict, test, and revise three times

Prediction before Run	Observed result	One code change	Result after change
Test 1			
Test 2			
Test 3			

Revise the matching pseudocode after each test

Test	Pseudocode line before	Revised pseudocode line	Evidence for the revision
1			
2			
3			

Improvement claim: Choose one test. When I changed ____, the program ____ because ____.

Checkpoint 4 - Plan and test one game feature

Feature goal: Name one challenge, reward, rule, identity choice, feedback cue, or progression feature.

Trigger and result: WHEN _____ happens, the program SHOULD _____.

Feature pseudocode: Write the feature logic before changing the game.

Test the planned feature

Prediction	Observed result	One change	Result after change
What should the feature do?	What did it do?	What exact block/value changed?	What improved?

Test claim: The feature changed ____ when _____. My evidence is _____.

Checkpoint 5 - Trace and repair the boundary bug

Do not repair the code until you record the prediction and first result.

Code element	Type	Purpose	Predicted value
missionName			
rows / columns			
priorityMode			
suppliesPlaced			

Prediction before Run: For rows = 3 and columns = 4, predict the grid and suppliesPlaced.

First observed result: Record the grid, score, and screenshot filename before repair.

Bugged inner-loop condition	Repaired inner-loop condition
column < _____	column < _____

After repair: Record the corrected grid, score, and screenshot filename.

Nested-loop explanation: The outer loop solves ____ because _____. The inner loop solves ____ because ____.

Checkpoint 6 - Create and improve a supply-grid program

Plan the counters and stop values here. In JavaScript, you must author the declarations and both complete nested-loop structures yourself.

Problem and user: Who needs this grid, and what should it help them organize?

Constraints: Choose 4 x 5 or 2 x 6. Keep every marker on screen and keep the score accurate.

Pseudocode before JavaScript

```
SET mission details
FOR each row
  FOR each column
    PLACE one supply
    UPDATE the count
  END FOR
END FOR
CHECK the mission result
```

Student-authored variables and operations

My JavaScript declaration	Type	Purpose	Operation I will use
	string		join text with +
	number		multiply / add
	Boolean		AND / comparison
Outer loop plan		Inner loop plan	
counter: row; stop: _____		counter: column; stop: _____	

Full-code evidence: Your submitted JavaScript must show both complete nested-loop structures and their bodies.

Checkpoint 6 continued - Test and revise

Prediction before Run: Predict totalExpected, suppliesPlaced, and one x/y position.

First observed result	Required revision	Result after revision	Evidence filename

Operations: Name one string, number, and Boolean operation visible in your code.

Subproblem explanation: Explain how the outer and inner loops solve different jobs in the supply-grid problem.

Improvement claim: The required revision improved the program because...

Checkpoint 7 - Plan the RVR mission

Evidence boundary

Use this page for problem solving and team planning. Use the existing RVR Mission Sheet for the sketch, robot ID, first-run result, revision, final photo, and final program evidence.

Mission problem: What recognizable word or symbol must the robot draw? What makes it challenging?

Possible solution	Programming mode / approach	Benefit	Limitation
Solution A			
Solution B			

Selected solution: Which solution will your team use, and why does it fit the problem better?

Team roles and timeline

Role	Student	Responsibility	Evidence due
Planner			
Programmer			
Tester / evidence lead			
Milestone	Expected time	Done?	Adjustment
Plan approved			
First run			
Revision run			

Checkpoint 7 continued - Robot pseudocode

Write the plan: Include headings, speed, duration/distance, waits, and repeated actions. Keep sketch, test, and final evidence on the existing RVR Mission Sheet.
